

Practice Quiz: Inventive Cognition

Part A: Multiple Choice

1. Analogical reasoning in invention works by: A) Copying the surface features of a solution from one domain to another B) Transferring structural relationships from a well-understood source domain to a less-understood target domain C) Randomly searching through unrelated domains until something fits D) Asking experts from different fields to collaborate
2. Gutenberg's printing press analogy from the wine press succeeded because: A) Wine presses and printing presses use the same materials B) Both devices are approximately the same size C) Both devices share the structural principle of uniform pressure applied across a surface D) Gutenberg was a winemaker before becoming a printer
3. According to Boden, productive combinatorial creativity requires: A) Combining as many components as possible for maximum feature richness B) Complementarity, emergence, and coherence among combined components C) Using only components from a single domain to ensure compatibility D) Patenting each component separately before combining them
4. In TRIZ, a "contradiction" occurs when: A) Two inventors disagree about the best solution B) The patent office rejects an application C) Improving one desirable parameter of a system worsens another desirable parameter D) A solution works in theory but fails in practice
5. Mental simulation in Active Inference corresponds to: A) Daydreaming about the future without systematic analysis B) Counterfactual evaluation of policies — predicting outcomes of actions not yet taken C) Physical prototyping at reduced scale D) Computer-aided design and finite element analysis
6. The SCAMPER method supports cognitive fluency by: A) Providing seven specific model-transformation operations that generate solution variants B) Eliminating all but

the best solution as quickly as possible C) Testing prototypes in seven different environments D) Surveying seven different user groups for feedback

7. Samsung's development of foldable OLED displays resolved the size-portability contradiction using which TRIZ principle? A) Segmentation — dividing the display into separate panels B) Dynamism — making the rigid display flexible and reconfigurable C) Inversion — making the display project outward instead of lying flat D) Nesting — placing one display inside another

Part B: Short Answer / Analysis

1. A medical device company needs to design a syringe that is both easy to use (requires low force) AND delivers medication precisely (requires high force for controlled flow). This is a classic TRIZ contradiction. Identify the contradicting parameters, then apply two different TRIZ principles to propose resolutions. Explain how each principle transforms the generative model of the syringe to resolve the contradiction.
2. The Wright brothers studied birds to understand flight control, eventually inventing wing warping. Analyze this as analogical reasoning using the structure-mapping framework. What structural relationships transferred from bird flight to airplane design? What surface features did NOT transfer? Identify one way the analogy could have gone wrong (a false mapping) and explain what evidence helped the Wrights avoid this failure.
3. You are developing an invention to reduce food waste in restaurant kitchens. Apply three different cognitive strategies from this module: (a) analogical reasoning — find a structural analogy from a non-food domain, (b) combinatorial creativity — combine two existing technologies into a novel solution, and (c) mental simulation — walk through a day-in-the-life scenario of a chef using your proposed solution. For each strategy, describe the specific concept it generates and assess its novelty and feasibility.